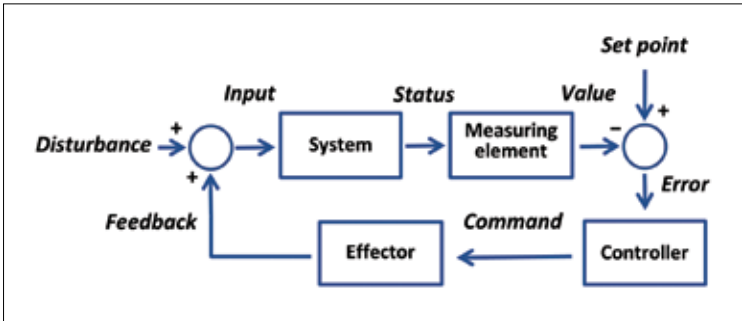




One of the standard models for implementing a control system

current, voltage, temperature, proximity and optical sensors. However, to be able to effectively use tactile sensors or chemical sensors may require some additional installations onto the processor. The development of such software has become a major focus of modern sensor research.

Further, for multipurpose robots, it becomes necessary to integrate the information provided by various sensors into an aggregated useful form. For instance, there may be a correlation between optical sensor data and measured temperature that proves more useful than either of these measurements alone - consider the example of a welding robot. For such cases, machine learning techniques for high-dimensional data may be used. The principal component analysis algorithm is one such useful algorithm. It works by finding the multidimensional 'directions' of the data that have maximum variance, hence identifying these as the unique directions that determine useful correlations in the data.

A classic example of the use of many kinds of sensors in one application, needing the regulation of well-designed software would be a quadcopter. Accelerometer data, gyroscope data, compass information, optical sensing, proximity sensors and perhaps even more advanced data like geopositioning information may need to be integrated to find the best possible combination of parameters to input for a desired output. The processing and complex mathematics made possible by software are critical to such nuanced predictions and direction.

The three laws of robotics

One of the pioneering thinkers of robotics, Isaac Asimov devised three rules for any robot to follow in order to ensure safe behaviour. These rules are-